

KALAKAN, Russian FederationWMO: **304690**Lat: **55.117N**Lon: **116.767E**Elev: **2011**StdP: **13.66**Time Zone: **9.00 (E09)**Period: **82-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-52.1	-47.5	-55.5	0.2	-51.6	-50.9	0.3	-47.1	10.0	-4.8	7.7	-6.6	0.0	20	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	82.5	62.9	79.2	61.7	75.9	60.2	67.0	76.5	64.9	73.6	62.9	71.2	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	96.0	69.7	61.9	89.0	67.9	59.8	82.7	66.5	32.8	76.0	31.1	73.9	29.6	70.8	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	9.8	7.8	DB	-55.9	88.7	6.4	2.9	-60.5	90.8	-64.2	92.5	-67.8	94.1	-72.5	96.2	(4)
(5)			WB	-55.9	72.4	6.4	4.4	-60.5	75.5	-64.2	78.1	-67.8	80.5	-72.4	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	-27.5	-13.9	4.1	27.4	43.2	57.0	62.2	57.4	43.0	23.4	-5.4	-24.3	(6)	
	DBStd	32.86	10.28	9.48	11.39	9.18	7.64	6.31	4.85	5.58	7.69	10.13	12.94	9.77	(7)	
	HDD50	11580	2401	1789	1422	679	238	14	1	9	237	824	1662	2304	(8)	
	HDD65	16198	2866	2209	1887	1129	677	248	111	240	661	1289	2112	2769	(9)	
	CDD50	894	0	0	0	0	26	224	379	239	26	0	0	0	(10)	
	CDD65	37	0	0	0	0	0	8	24	5	0	0	0	0	(11)	
	CDH74	1075	0	0	0	0	43	369	504	155	4	0	0	0	(12)	
	CDH80	214	0	0	0	0	5	81	106	22	0	0	0	0	(13)	
	WSAvg	1.6	0.4	0.7	1.8	3.3	3.3	2.1	1.6	1.4	1.8	1.8	0.9	0.3	(14)	
	PrecAvg	15.6	0.1	0.1	0.2	0.7	1.3	2.7	3.7	3.4	2.0	0.7	0.3	0.2	(15)	
(16) Precipitation	PrecMax	23.4	0.4	0.6	0.6	2.8	3.7	5.7	9.7	6.7	5.6	3.1	0.7	0.7	(16)	
	PrecMin	9.2	0.0	0.0	0.0	0.0	0.3	0.5	0.6	0.7	0.2	0.0	0.0	0.0	(17)	
	PrecStd	3.1	0.1	0.2	0.2	0.6	0.8	1.5	1.8	1.6	1.2	0.6	0.2	0.2	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	22.9	44.7	60.2	79.6	86.5	87.7	82.9	73.7	53.8	28.7	10.2	(19)	
		MCWB	4.8	18.3	35.5	45.2	55.9	63.7	66.4	64.0	57.3	42.1	24.3	8.3	(20)	
	2%	DB	-0.9	15.7	36.9	52.7	73.2	82.2	83.2	78.3	67.5	48.1	23.0	1.5	(21)	
		MCWB	-2.2	12.5	29.2	40.1	52.9	61.0	64.7	62.1	55.1	37.7	20.0	0.2	(22)	
	5%	DB	-5.7	10.7	31.9	47.9	67.6	78.6	79.7	74.4	62.2	43.3	18.2	-4.6	(23)	
		MCWB	-6.7	8.2	25.7	37.2	50.2	59.8	63.4	60.4	51.0	34.9	16.2	-5.2	(24)	
	10%	DB	-10.8	6.1	27.0	43.7	61.9	74.4	76.4	70.8	57.5	39.0	13.9	-9.3	(25)	
		MCWB	-11.3	4.0	22.0	34.4	47.4	57.8	62.2	59.3	48.1	32.2	12.3	-9.8	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	18.7	35.9	45.8	57.9	68.0	70.2	68.2	60.6	42.8	25.3	8.7	(27)	
		MCDB	6.0	23.1	44.8	57.9	76.3	81.7	79.7	75.4	67.6	52.9	28.0	10.0	(28)	
	2%	WB	-2.0	12.7	29.7	41.0	54.2	63.9	67.8	64.9	56.4	38.7	20.2	0.4	(29)	
		MCDB	-0.8	15.7	36.3	51.4	69.3	75.3	77.6	72.9	64.3	46.8	22.7	1.4	(30)	
	5%	WB	-6.6	8.3	26.0	37.9	51.2	61.6	66.1	62.9	53.3	35.7	16.1	-5.3	(31)	
		MCDB	-5.8	10.7	31.9	46.8	64.7	73.3	74.9	70.2	60.7	41.7	18.1	-4.7	(32)	
	10%	WB	-11.3	4.1	22.0	35.2	48.3	59.5	64.4	60.8	49.6	32.9	12.4	-9.8	(33)	
		MCDB	-10.7	6.2	26.9	42.8	60.8	70.6	72.9	67.9	56.3	38.4	14.1	-9.3	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	24.0	36.5	38.6	27.8	26.7	27.7	25.7	24.8	24.7	25.6	25.0	21.1	(35)	
		MCDBR	27.2	39.4	39.8	33.5	40.0	38.4	33.7	32.7	31.9	29.4	26.7	23.3	(36)	
		MCWBR	26.0	36.0	32.2	21.9	20.9	18.0	16.4	17.2	18.2	19.3	23.5	22.3	(37)	
	5% WB	MCDBR	26.8	39.5	38.9	31.4	36.3	30.9	26.2	25.9	29.0	27.5	26.2	23.2	(38)	
		MCWBR	25.6	36.2	31.5	21.2	20.2	16.2	14.8	15.3	17.3	18.5	23.0	22.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.244	0.276	0.306	0.388	0.482	0.467	0.441	0.405	0.343	0.299	0.264	0.242	(40)	
	taud		2.114	2.096	2.101	2.064	1.920	2.091	2.211	2.294	2.416	2.403	2.186	2.062	(41)	
	Ebn at Noon		218	252	272	262	241	248	253	256	260	246	207	184	(42)	
	Edh at Noon		20	29	37	45	56	48	42	36	28	22	18	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		329	678	1237	1701	1696	1822	1704	1442	1033	682	381	239	(44)	
	RadStd		10	14	29	81	118	104	140	97	79	39	20	8	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page